Hazardous Waste Impact on Low Birth Weights

Analyzing the correlation between infancy weights near hazardous waste sites in California

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Background

Our why,

- Unequal Human Health Impacts
- Environmental Justice
- Inform Policy Development
- Public Health Risks

Research Question + Goals

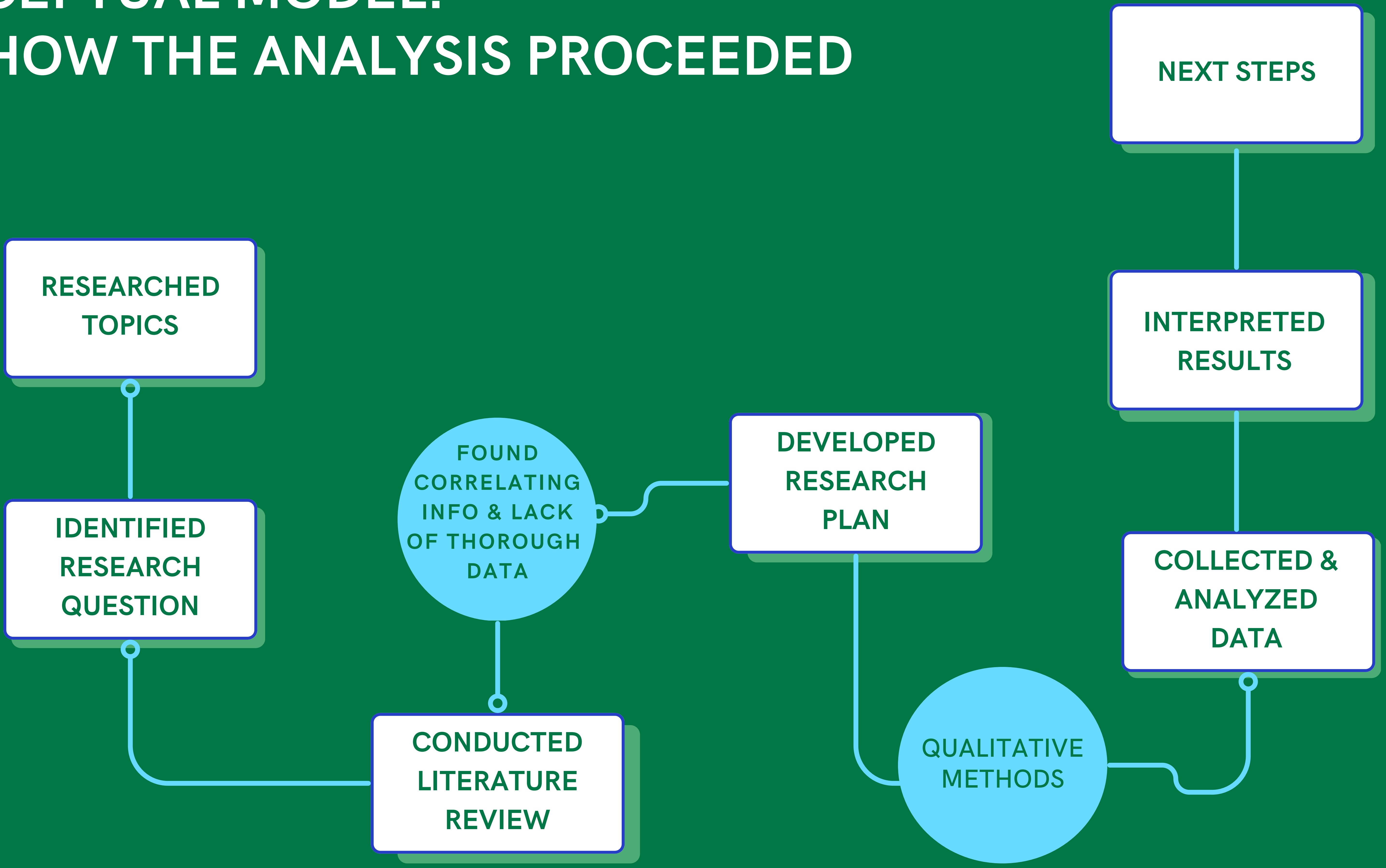
Does close proximity to hazardous waste sites correlate with the percent occurrence of low birth weights?

R Studio for Linear Models

R Studio for Summary Statistics

ArcGIS

CONCEPTUAL MODEL:
FOR HOW THE ANALYSIS PROCEEDED

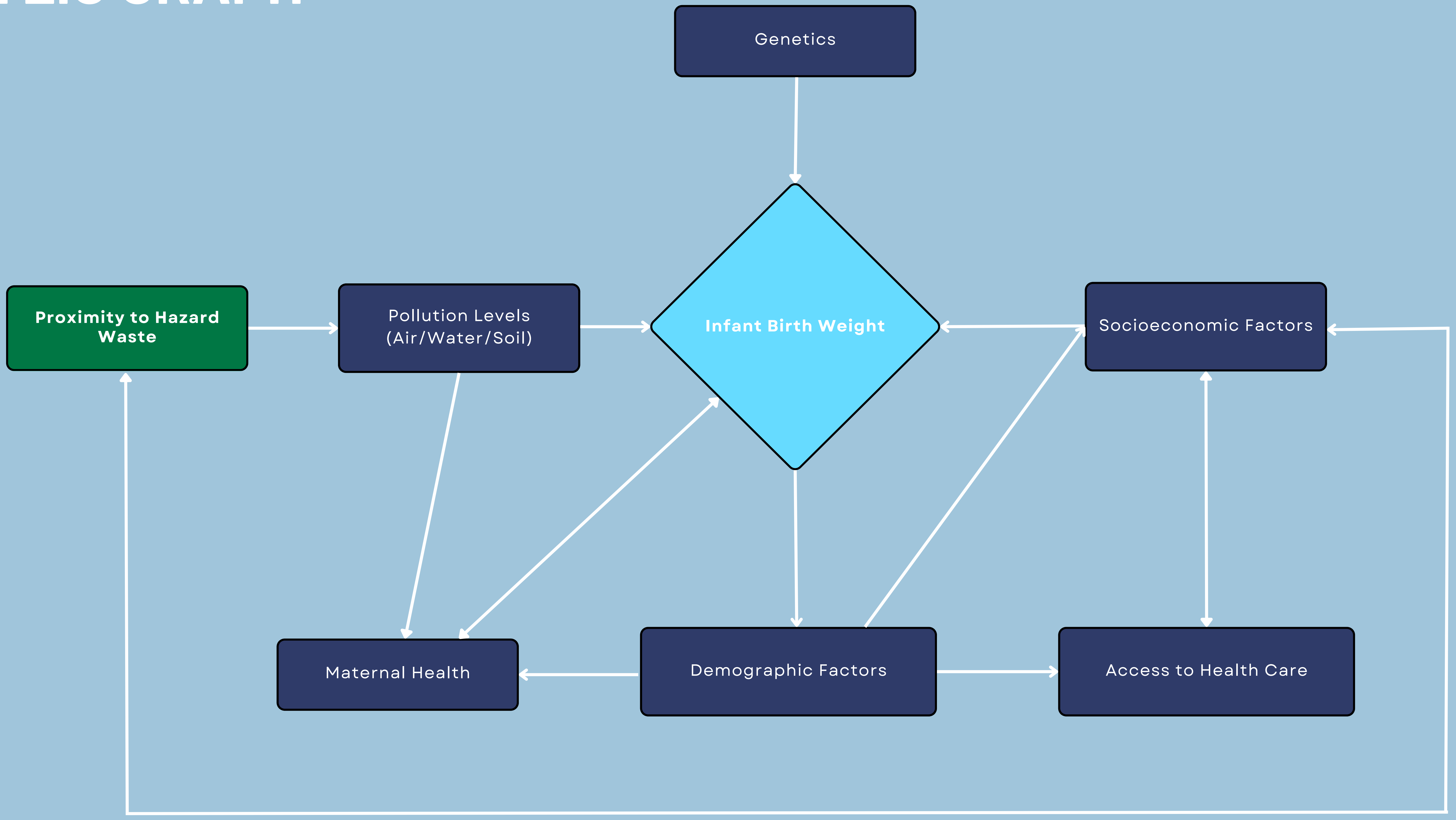


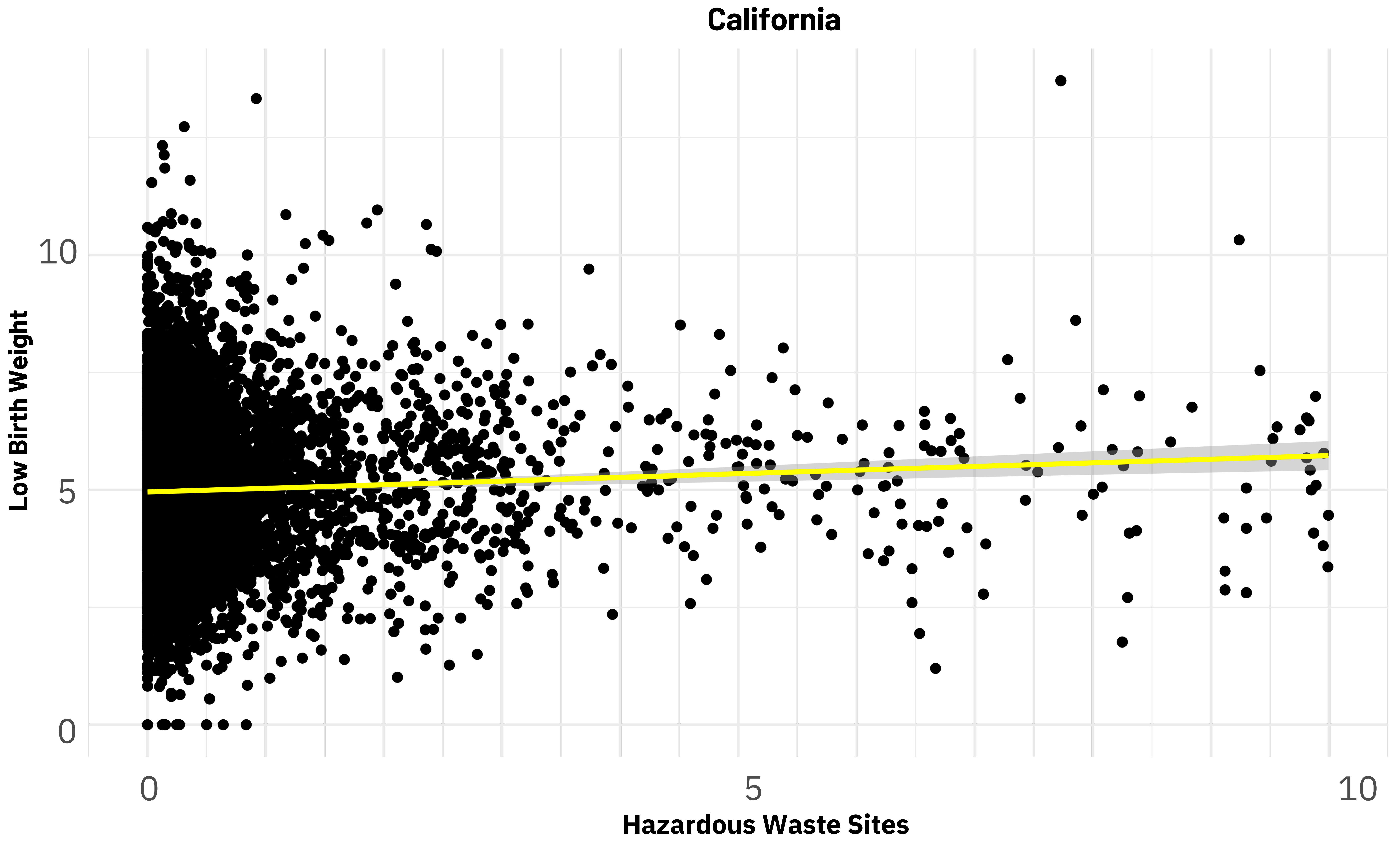
DATA SOURCES

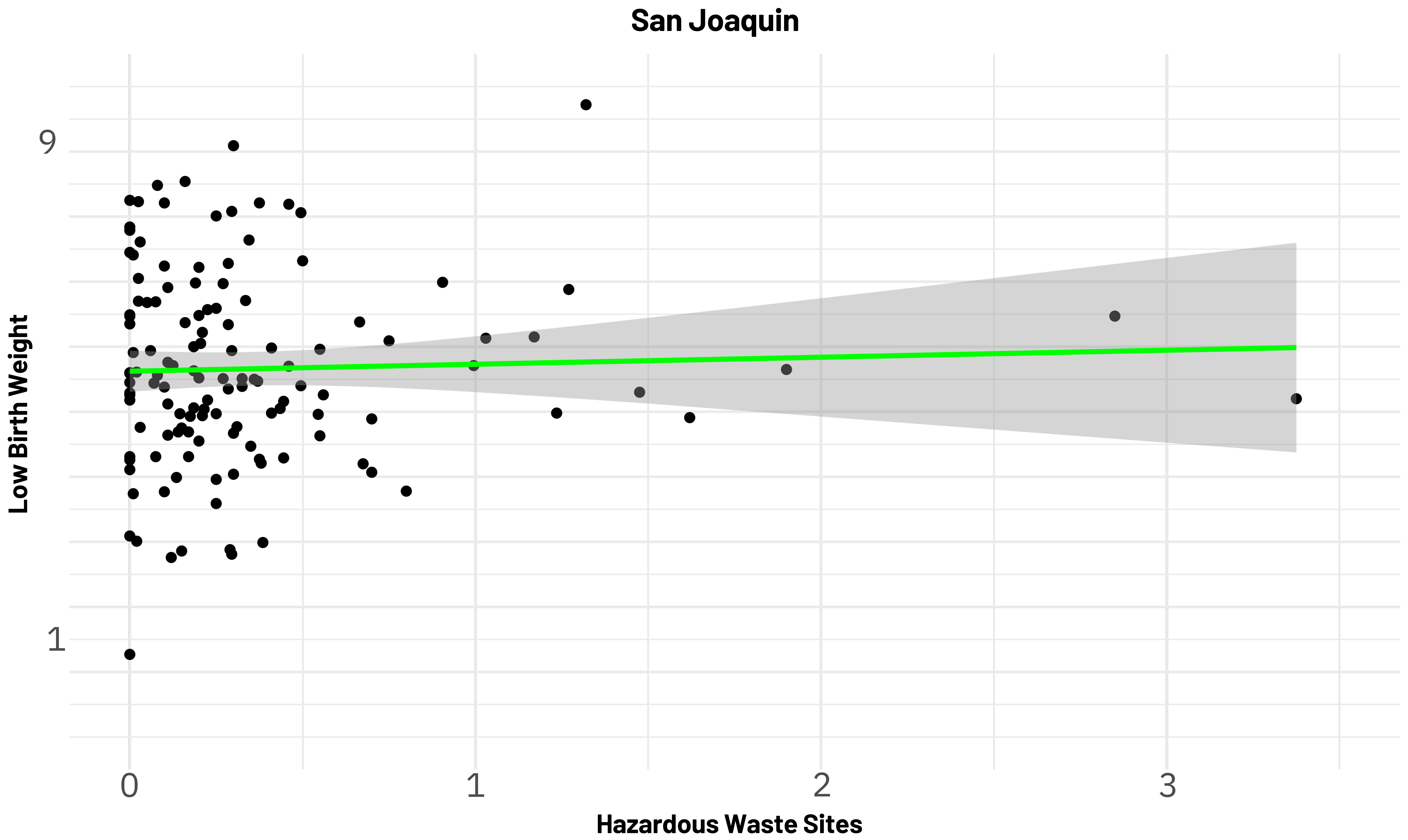
Cal Enviro Screen
4.0

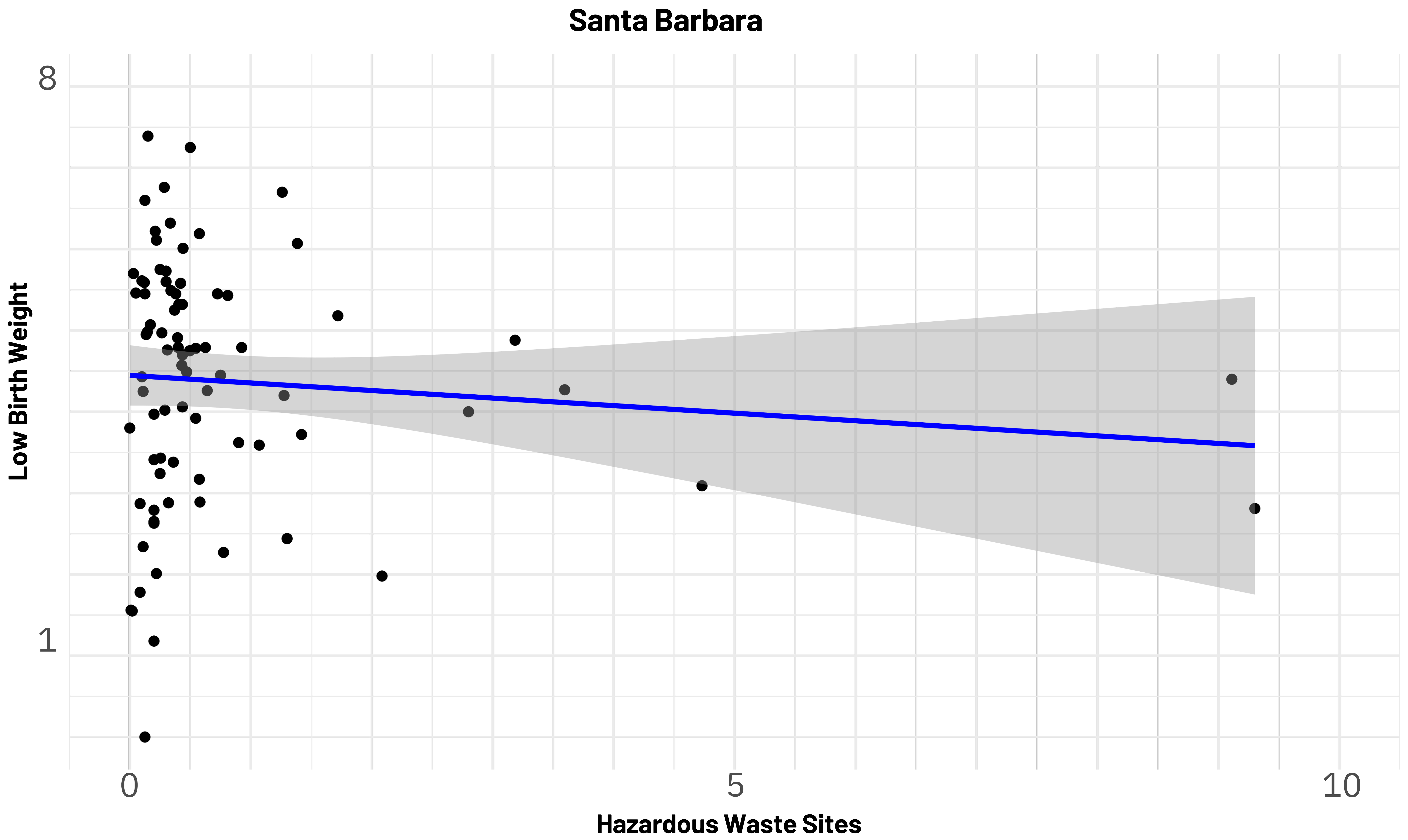
U.S. Census Tract

CONCEPTUAL MODEL: DIRECTED ACYLIC GRAPH

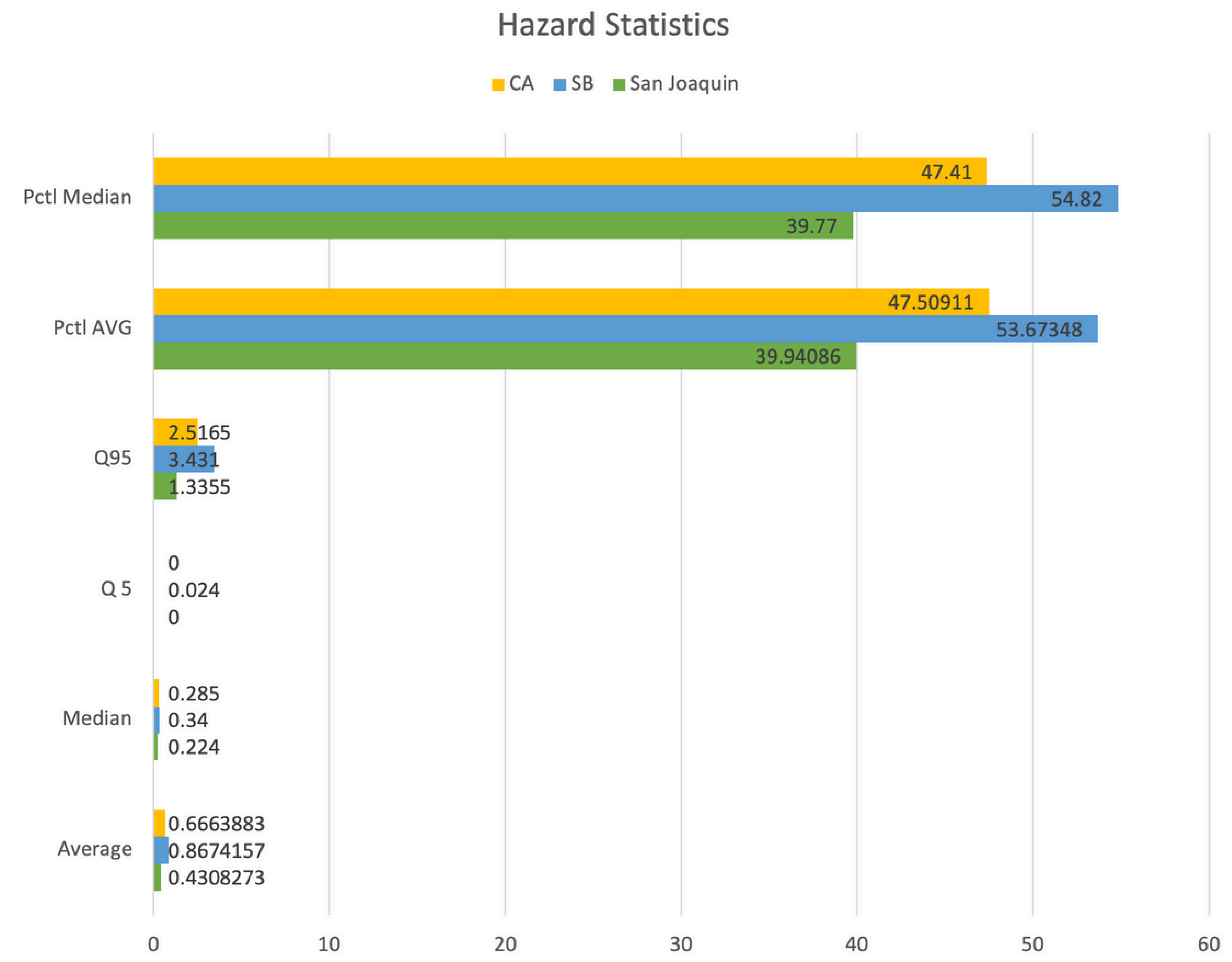



California
Low Birth Weight
Hazardous Waste Sites
0
5
10
0
5
10

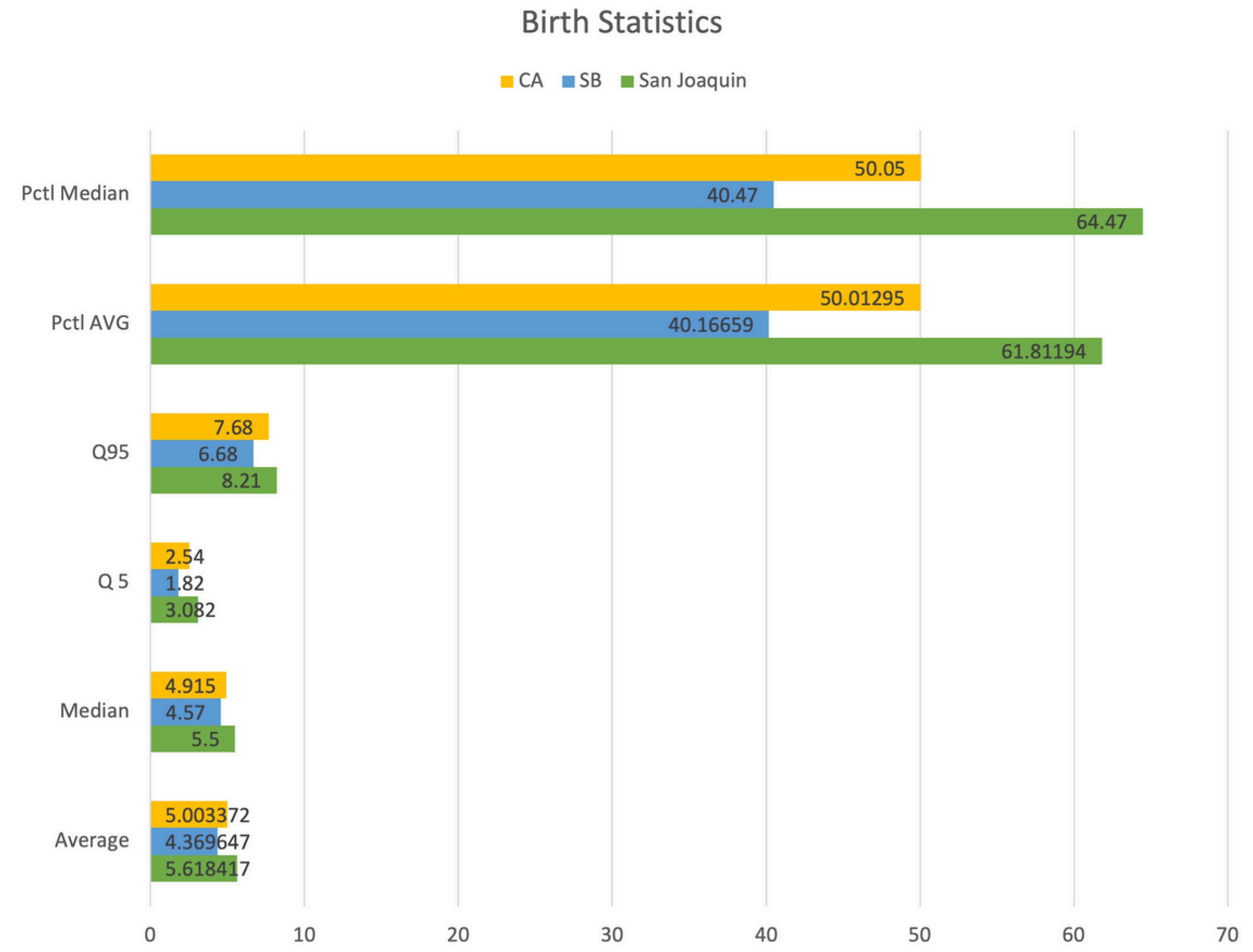

San Joaquin
9
1
Low Birth Weight
0
1
2
3
Hazardous Waste Sites


Santa Barbara
8
1
0
5
10
Low Birth Weight
Hazardous Waste Sites

STATISTICAL DATA

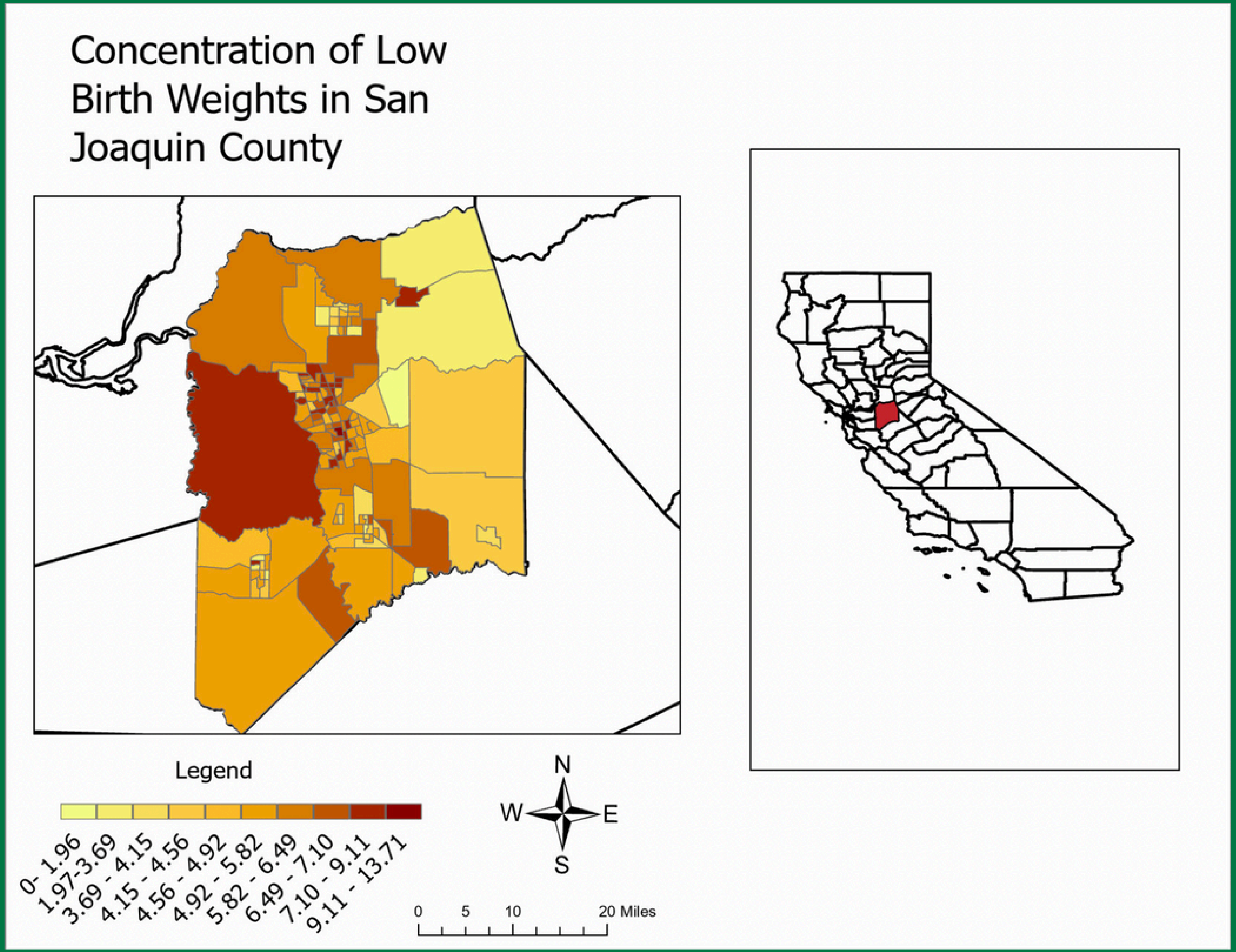


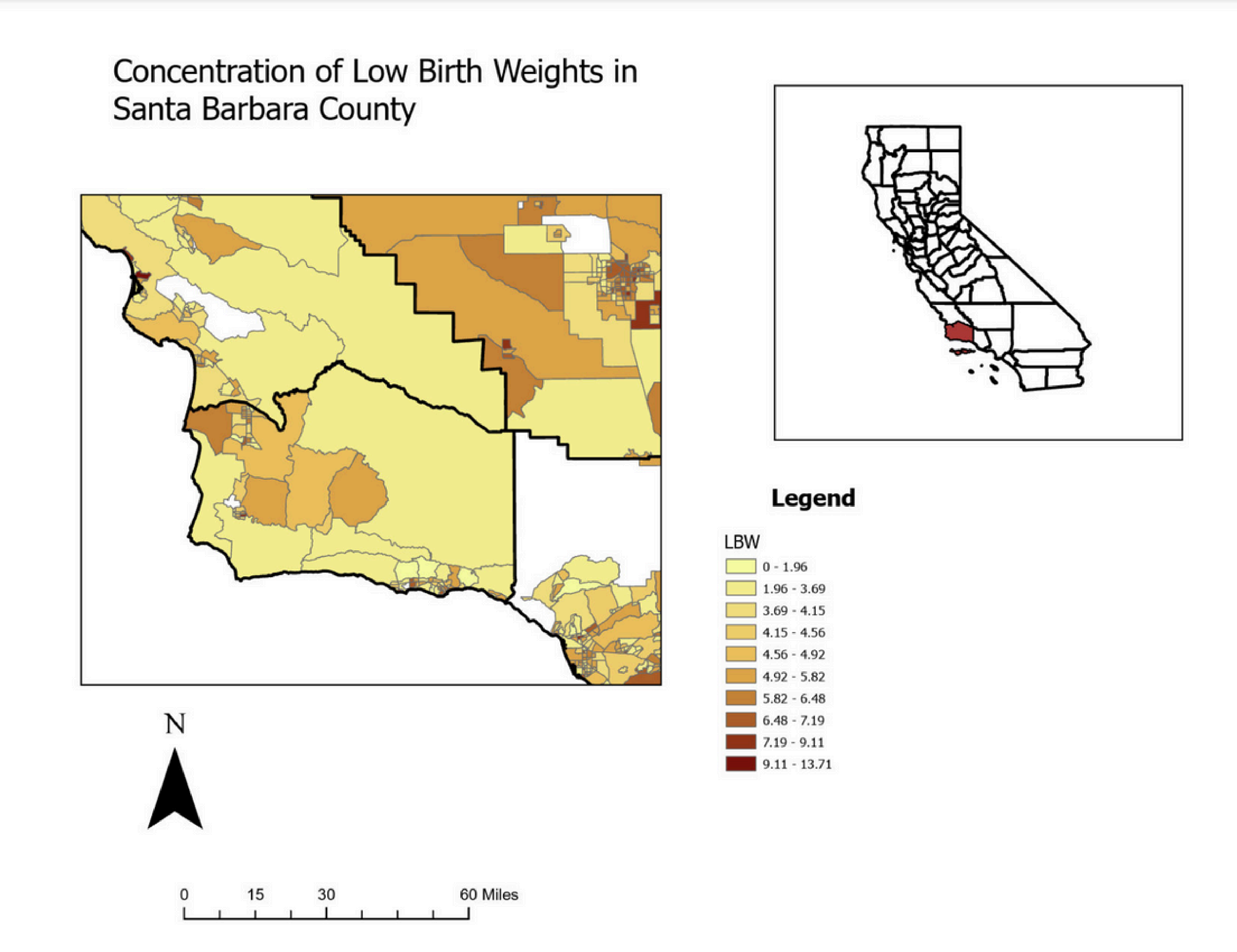
	Average	Median	Q 5	Q95	Pctl AVG	Pctl Median
CA	0.6663883	0.285	0	2.5165	47.50911	47.41
SB	0.8674157	0.34	0.024	3.431	53.67348	54.82
San Joaquin	0.4308273	0.224	0	1.3355	39.94086	39.77



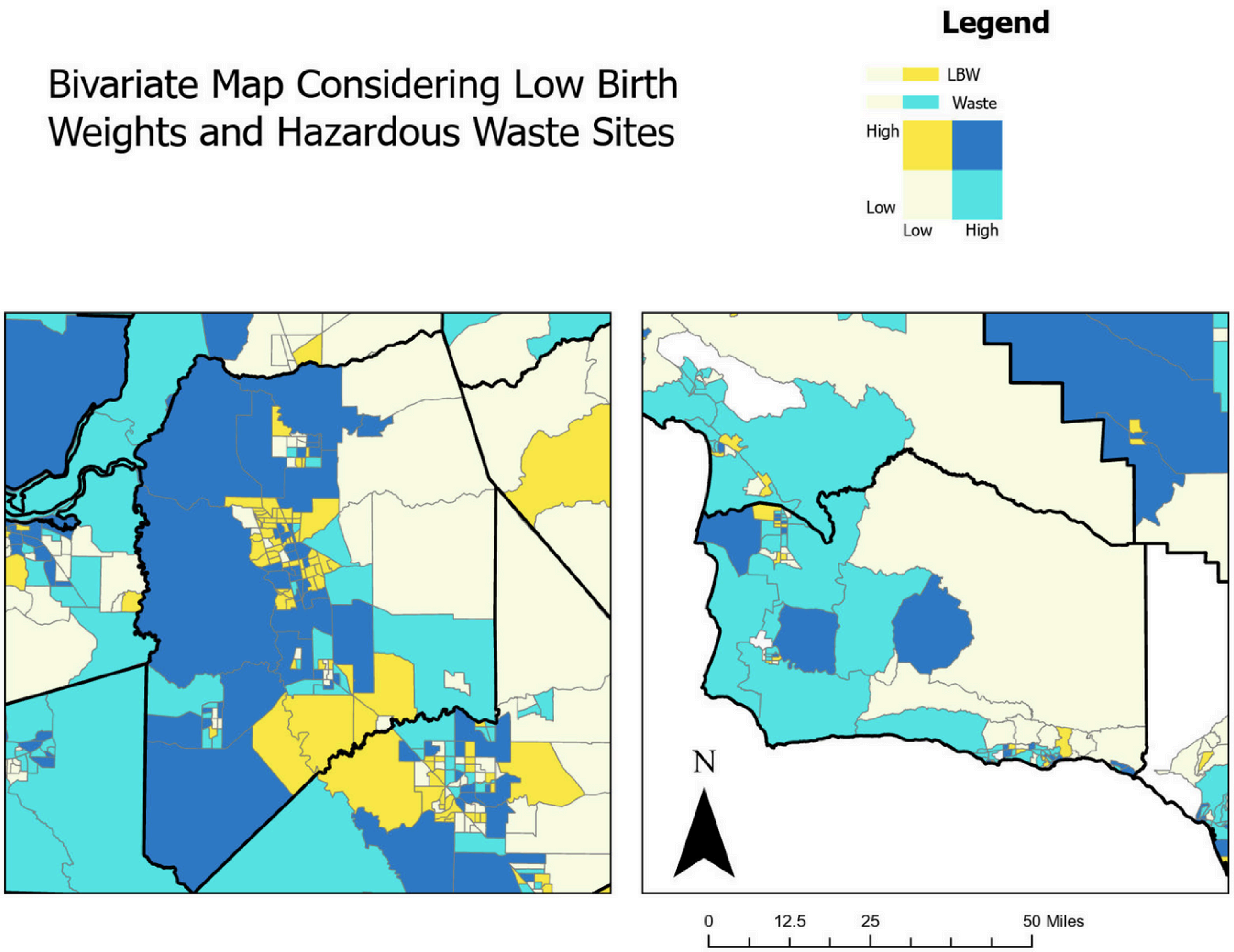
	Average	Median	Q 5	Q95	Pctl AVG	Pctl Median
CA	5.003372	4.915	2.54	7.68	50.01295	50.05
SB	4.369647	4.57	1.82	6.68	40.16659	40.47
San Joaquin	5.618417	5.5	3.082	8.21	61.81194	64.47

COMPARATIVE SPACIAL DATA





Bivariate Map Considering Low Birth Weights and Hazardous Waste Sites



RESULTS

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We found that there is a relationship between Hazardous Waste sites and the percent occurence of low birth weights

→

We found that San Joaquin and Santa Barbara were affected differently in terms of low birth weight occurrence

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With more time and resources, we would further investigate this topic and improve upon our analysis

FUTURE DIRECTIONS

- INCORPORATE MORE VARIABLES SUCH AS:
  - Access to healthcare
  - Medi-Cal Data
  - Groundwater Contamination
  - Long term effects of LBW
- Look Deeper into San Joaquin
- Research if Socioeconomic factors in counties influence how much hazardous waste affects low birth weights

CITATIONS & THANK YOU!

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- Chernausek, S. D. (2012). UPDATE: Consequences of abnormal fetal growth. The Journal of Clinical Endocrinology & Metabolism, 97(3), 689–695. https://doi.org/10.1210/jc.2011-2741

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